



13.2.5 Wrap-Up: Ticket Out the Door

- Complete the table to compare two investments.

	Simple Interest	Compound Interest
Investment Amount and APR	\$2,325 is invested at an interest rate of 4.5% APR	\$2,325 is invested at an interest rate of 4.5% APR compounded monthly
Write the equation.	$A = P(1 + rt)$	$A = P \left(1 + \frac{r}{n}\right)^{nt}$
Determine the amount after 8 years.		

Unit 13, Lesson 3: Comparing Key Features of Functions



Benchmarks

MA.912.F.1.6 Compare key features of linear and nonlinear functions each represented algebraically, graphically, in tables, or written descriptions.

MA.912.F.1.8 Determine whether a linear, quadratic, or exponential function best models a given real-world situation.



Learning Targets

- ☐ I can compare key features of linear and nonlinear functions that are represented algebraically, graphically, in tables, or in written descriptions.
- ☐ I can verify when a quantity increasing exponentially exceeds a quantity increasing linearly or quadratically.



13.3.1 Warm-Up: Tale of Three Patterns

1. The table shows three different patterns.

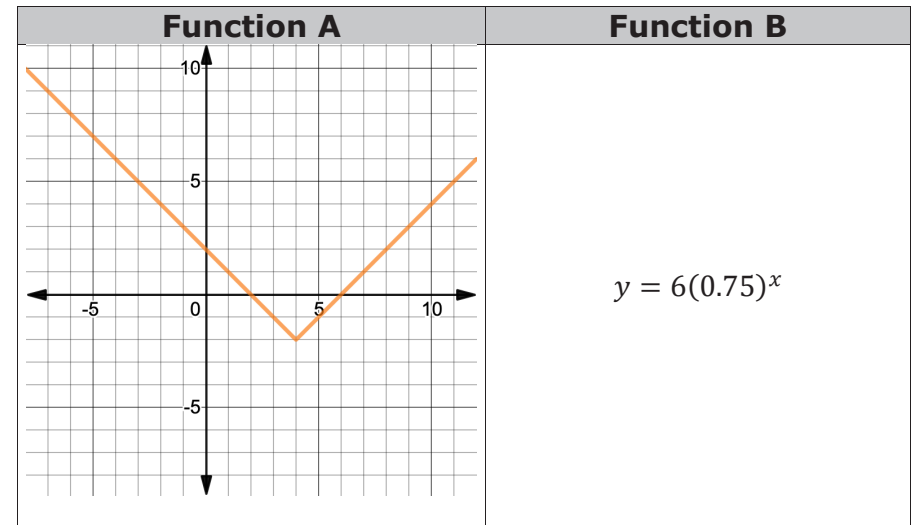
Step	Pattern A	Pattern B	Pattern C
1			
2			
3			
4			
5			

- Write the number of squares needed for step 5 of each pattern.
- Which pattern increases the same number of squares for each step?



13.3.2 Guided Instruction: Comparing Functions

1. The table shows two functions.

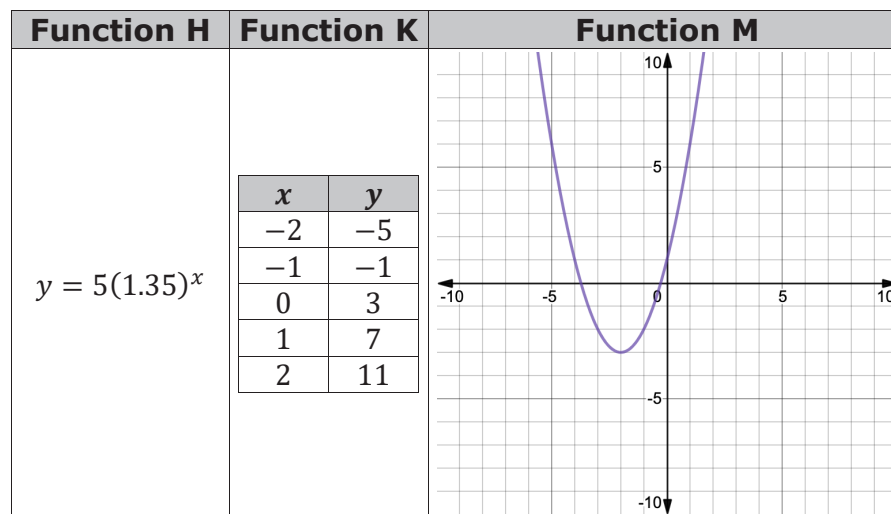


- Which function has the smallest y -intercept?
 - ☐ Function A
 - ☐ Function B
- As $x \rightarrow \infty$, which function has larger values of y ?
 - ☐ Function A
 - ☐ Function B

- c. Compare the ranges of the functions and indicate any similarities or differences.

- d. On the interval where Function A is negative, what type of values does Function B have?

2. The table shows three different functions.



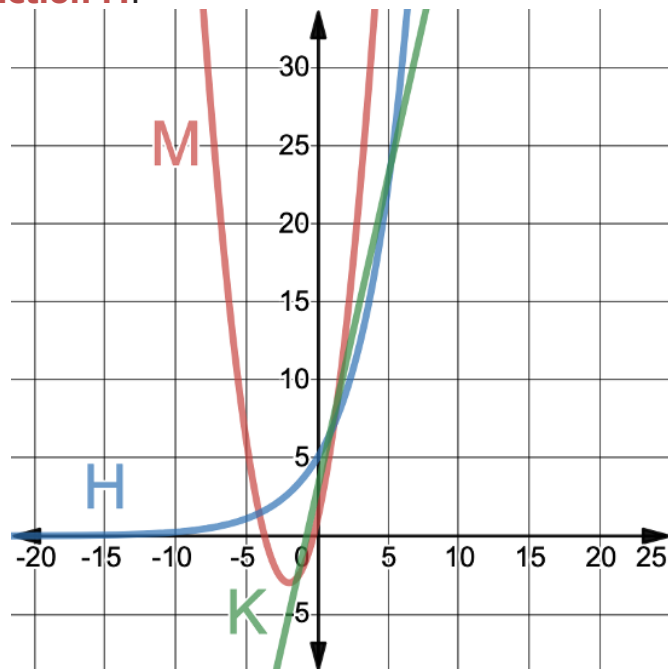
- a. Determine the type of function that each represents.

Function H	Function K	Function M
☐ linear ☐ quadratic ☐ exponential	☐ linear ☐ quadratic ☐ exponential	☐ linear ☐ quadratic ☐ exponential

- b. Which key features are the same for all three functions?

- ☐ domain
- ☐ range
- ☐ x -intercepts
- ☐ y -intercepts
- ☐ right end behavior
- ☐ left end behavior
- ☐ intervals where the function is positive
- ☐ intervals where the function is negative
- ☐ intervals where the function is increasing
- ☐ intervals where the function is decreasing

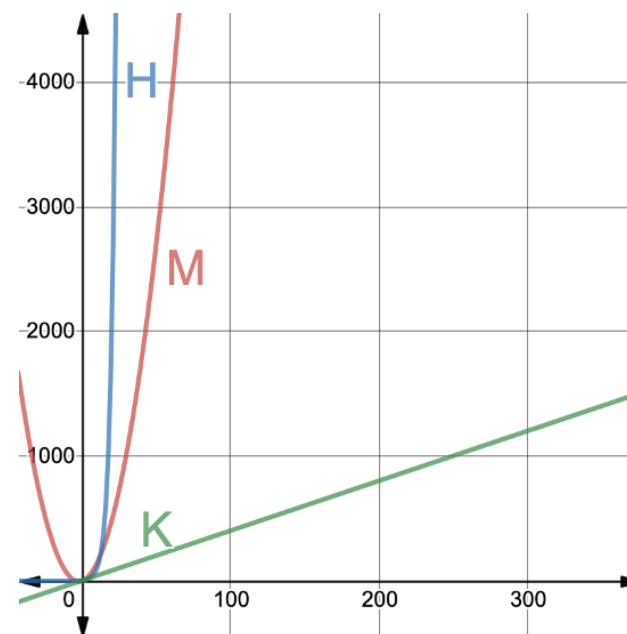
3. The graph shows **Function H**, **Function K**, and **Function M**.



Which function, **H**, **K**, or **M**, will eventually exceed the other functions as all the functions increase?

- ☐ Function H
- ☐ Function K
- ☐ Function M

4. The graph shows **Function H**, **Function K**, and **Function M** using different scales for the axes.



Discuss with your partner whether or not the graph verifies that the function you chose in question 3 will eventually exceed the other functions as all the functions increase. If necessary, adjust your answer.

5. Comparing simple interest to compound interest is like comparing a linear function to an exponential function. As observed in the previous lesson, the amount of interest accrued with compound interest exceeds simple interest. In the graphs of functions **H**, **K**, and **M**, the quadratic function, **M**, rises faster than the linear function, **K**, but it does not rise faster than the exponential function **H**.

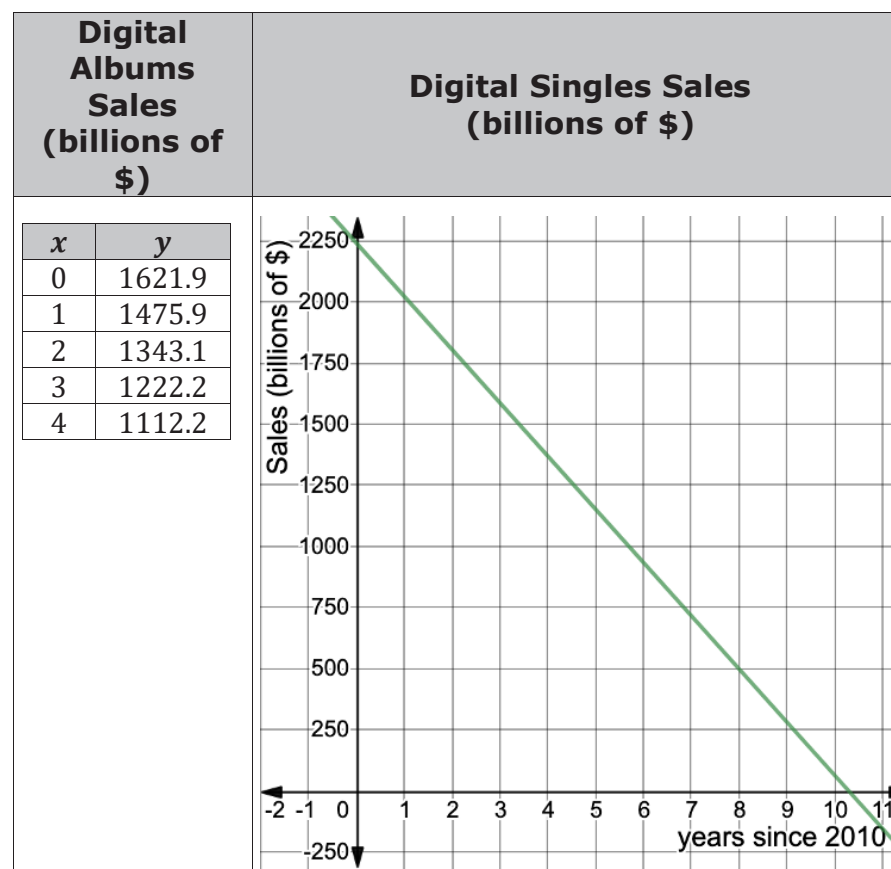
Discuss with your partner why exponential growth will always eventually exceed a parabola that opens up, and a line with a positive slope. Summarize your discussion.



13.3.3 Collaboration: Digital Music Sales

Work with your partner to complete the following.

- The table shows a table of values representing the revenue, in billions of dollars, from sales of digital albums, where $x = 0$ is 2010, and a graph representing the revenue, in billions of dollars, from sales of digital singles, where $x = 0$ is 2010.



- a. Approximate the x -intercept for the function that represents sales of digital singles.
- b. Interpret the x -intercept of the function that represents sales of digital singles.
- c. Discuss with your partner whether or not the interpretation is accurate. Summarize your discussion.
- d. What is the domain of the function representing sales of digital singles?

- e. Work with your partner to determine the key features for each function.

Key Feature	Digital Album Sales	Digital Single Sales
y -intercept		
End Behavior	As $x \rightarrow -\infty$, $y \rightarrow$ _____ As $x \rightarrow \infty$, $y \rightarrow$ _____	As $x \rightarrow -\infty$, $y \rightarrow$ _____ As $x \rightarrow \infty$, $y \rightarrow$ _____
Increasing or Decreasing	The function is _____ over time.	The function is _____ over time.

- f. Discuss with your partner what type of function best describes the table of values that represents the digital album sales. Explain your choice.
 - linear
 - quadratic
 - exponential

2. The table shows written descriptions and equations that model the home prices, in thousands of dollars, for two different time periods.

Home Prices in Florida	
2009-2014	2015-2019
The average price for a house in Florida between 2009 and 2014, y , in thousands of dollars, is modeled by the equation $y = 4.4x^2 - 22.1x + 160.8$, where x is the number of years since 2009.	In 2015, the average home price in Florida was 175.6 thousand dollars. For each year between 2015 and 2019, the average price increased by 19.1 thousand dollars.

- a. Work with your partner to determine the key features. Give the domain in set-builder notation.

Key Feature	Home Prices in Florida 2009-2014	Home Prices in Florida 2015-2019
Domain		
y -intercept		

- b. Which context has the larger y -intercept?

- c. From standard form, the vertex of the parabola can be found using $\left(\frac{-b}{2a}, c - \frac{b^2}{4a}\right)$. Determine the vertex of the parabola that models home prices in Florida from 2009 to 2014.

- d. Use the domain and the vertex to determine the range for home prices in Florida from 2009 to 2014.

- e. Determine the type of function that models the home prices for 2015 to 2019.

- f. Use the information given for the home prices for 2015 to 2019 to determine the range of prices.

- g. Which period, 2009–2014 or 2015–2019, has the greatest range of prices?



13.3.4 Wrap-Up: Reflection

1. Complete the table.

The terms and information used in the Guided Instruction and Collaboration that I knew were ...	Something I wondered during the Guided Instruction or Collaboration was ...	Something I learned during the Guided Instruction or Collaboration was ...



Unit 13, Lesson 4: Parent Functions



Benchmark

MA.912.F.1.1 Given an equation or graph that defines a function, classify the function type. Given an input-output table, determine a function type that could represent it.



Learning Targets

- ☐ I can match multiple representations of the parent function to its table of values.
- ☐ I can determine a function type that could represent a transformation of a parent function.